

Position: Internship cum PPO

Duration: 24 Hours

Objective: Build a functional MVP of an AI-powered platform capable of autonomously discovering and managing editorial backlink acquisition opportunities.

Overview

We are building the next generation of autonomous AI systems. One of our flagship products is an AI agent capable of performing the work of an SEO outreach specialist from understanding a website to identifying backlink opportunities and generating personalized outreach.

This challenge is designed to evaluate your ability to rapidly build AI-first products, orchestrate multiple APIs, design scalable architectures, and deliver an end-to-end MVP under tight deadlines.

The expectation is **not a polished production product**, but a working demonstration of your engineering ability, system thinking, and execution speed.

Project Vision

Imagine a user simply enters:

<https://company.com>

The AI should automatically:

- Understand the business
- Understand its niche
- Analyze the website
- Discover backlink opportunities
- Rank opportunities
- Find contact information
- Generate personalized outreach emails
- Track outreach status
- Monitor acquired backlinks

The user should feel like they hired an AI SEO specialist.

Problem Statement

Acquiring quality backlinks is one of the most manual and time-consuming parts of SEO.

The current workflow involves:

- Finding relevant websites
- Checking authority
- Finding editors
- Finding emails
- Writing outreach
- Sending emails

- Following up
- Tracking responses
- Monitoring backlinks

This process takes dozens of hours every month.

Our objective is to automate as much of this workflow as possible using AI.

MVP Objective

Build an AI agent capable of acquiring editorial backlink opportunities.

The MVP should demonstrate the complete workflow, even if some parts use mock data or simulated integrations.

Core Workflow

User Enters Website URL



AI Understands the Business



AI Identifies Target Keywords



AI Discovers Relevant Websites



AI Qualifies Opportunities



AI Finds Decision Makers



AI Generates Personalized Outreach



AI Sends Outreach Campaign



AI Handles Replies



AI Generates Guest Article (when requested)



AI Delivers or Publishes Content (where authorized)



AI Confirms Backlink Placement



AI Monitors Backlink Health



AI Generates Campaign Report

Required Features

Functional Requirements

The AI Agent should autonomously perform the following tasks:

1. Website Intelligence

- Crawl the provided website
- Understand the business
- Identify industry
- Extract keywords
- Understand products/services
- Generate business summary

2. Opportunity Discovery

Discover backlink opportunities such as:

- Guest Post websites
- Resource Pages
- Startup Directories
- Business Directories
- Industry Blogs
- Editorial Publications
- Niche Communities
- Podcast opportunities
- Broken Link opportunities
- Unlinked Brand Mentions

3. Opportunity Qualification

Score every opportunity based on

- Domain Authority
- Organic Traffic
- Relevance
- Editorial Quality
- Spam Score
- Trust Score

Return only high-quality prospects.

4. Contact Discovery

Locate

- Editor
- Website Owner
- Marketing Manager
- SEO Manager

Find

- Email
- Contact Form
- LinkedIn
- Submission Guidelines

5. Personalized Outreach

Generate highly personalized outreach messages.

Every email should reference

- Prospect website
- Recent article
- Business relevance
- Value proposition

Generic outreach templates are not acceptable.

6. Outreach Automation

The AI should

- Send emails
- Schedule follow-ups
- Track opens
- Track replies
- Track status

Maintain a CRM for all outreach activity.

7. AI Reply Handling

Automatically classify incoming replies.

Examples

- Interested
- Rejected
- Question
- Need more information
- Guest post accepted
- Pricing discussion

Generate an appropriate draft response.

8. Editorial Content Generation

If a guest article or resource is requested, the AI should generate:

- SEO-optimized article
- Title
- Meta description
- Headings
- Internal linking suggestions
- External references
- FAQ section
- Schema markup

The content should be publication-ready.

9. Backlink Placement

Where the user has authorization (e.g., their own WordPress site, a connected CMS, or a partner site with API access), the AI should be capable of:

- Publishing the article
- Submitting the article
- Preparing the submission package
- Completing supported submission workflows

For third-party websites, the AI should prepare everything needed for editorial review rather than bypassing approval.

10. Backlink Verification

After publication, verify

- Link exists
- Correct URL
- Correct anchor text
- DoFollow / NoFollow
- HTTP Status
- Page indexed
- Link still active

11. Dashboard

Display

- Opportunities Found
- Qualified Prospects
- Outreach Sent
- Open Rate
- Reply Rate
- Accepted Opportunities
- Published Articles
- Live Backlinks
- Average Domain Authority
- Campaign Performance

Bonus Features

Exceptional candidates may also implement:

- LangGraph multi-agent orchestration
- AI planner that decides the best backlink strategy
- Prospect memory using a vector database
- Competitor backlink mining
- Digital PR workflow
- Broken-link replacement automation
- Unlinked brand mention detection
- Chrome extension
- Docker deployment

- Real-time analytics
- Agency multi-workspace support

Success Criteria

By the end of the challenge, we should be able to provide only a website URL and observe an AI system that:

- Understands the business.
- Finds relevant editorial backlink opportunities.
- Identifies the right contacts.
- Creates highly personalized outreach.
- Manages communication.
- Produces publication-ready content when needed.
- Supports publishing where authorized.
- Verifies acquired backlinks.
- Tracks campaign performance through a professional dashboard.

This demonstrates the ability to build an AI-native workflow that automates legitimate backlink acquisition while respecting editorial review and website permissions.

Technical Expectations

We expect clean architecture.

Suggested stack:

Frontend

- Next.js
- React
- Tailwind CSS
- shadcn/ui

Backend

- FastAPI

AI

- OpenAI GPT-5 (or another capable LLM)
- LangGraph or LangChain (preferred)

Database

- PostgreSQL
- SQLite (acceptable for MVP)

Crawler

- Playwright
- Firecrawl

Search

- SerpAPI
- Google Search APIs
- Mock dataset (acceptable)

Deployment

- Docker (preferred)
- Vercel + Railway/Fly.io also acceptable

Deliverables

Your submission should include:

Source Code

GitHub Repository

README

Include:

- Setup instructions
- Architecture overview
- Technologies used
- Assumptions
- Known limitations
- Future improvements

Demo Video

5–10 minutes demonstrating:

- Website analysis
- Opportunity discovery
- AI-generated outreach
- Dashboard walkthrough

Live Deployment

Deploy if possible (on vercel)

Evaluation Criteria

We are evaluating engineering ability not just features.

Product Thinking (20%)

- Does the workflow make sense?
- Is the user experience intuitive?
- Are reasonable product decisions made?

AI Integration (25%)

- Effective use of LLMs
- Quality of generated summaries
- Quality of personalized outreach
- Appropriate prompting and orchestration

Engineering Quality (20%)

- Code organization
- API design
- Separation of concerns
- Error handling
- Maintainability

Functionality (20%)

- End-to-end workflow works
- Website analysis functions
- Opportunities displayed
- Outreach generated

UI/UX (15%)

- Clean interface
- Responsive design
- Professional appearance
- Clear information hierarchy

Constraints

- Do **not** use black-hat SEO techniques.
- Do **not** automate posting or purchasing backlinks.
- Focus on identifying **editorial** backlink opportunities and generating high-quality, personalized outreach.
- If a third-party API is unavailable, clearly state assumptions and use mock data where necessary.

What We're Looking For

We are not expecting perfection in 24 hours.

We are looking for someone who can:

- Break down complex problems
- Build AI-first systems
- Learn quickly
- Deliver working software rapidly
- Write clean, maintainable code
- Make pragmatic engineering decisions under time constraints

Bonus Points

You will stand out if your solution includes:

- Multi-agent orchestration (e.g., LangGraph)
- Autonomous task planning
- Persistent memory for prospects
- Human-in-the-loop approval for outreach
- A polished, modern dashboard
- Dockerized deployment
- Comprehensive documentation

Final Goal

By the end of the challenge, we should be able to enter a website URL and watch the platform autonomously:

1. Analyze the business.
2. Discover relevant editorial backlink opportunities.
3. Prioritize the best prospects.
4. Generate personalized outreach.
5. Present everything through a clean dashboard.

The objective is to demonstrate that you can architect and build an AI-native MVP under real-world startup